

172 Tb/s GMI, O-band Coherent Transmission with OSNR Optimization

Aleksandr Donodin⁽¹⁾, Shekhar Saxena⁽¹⁾, Daniel J. Elson⁽²⁾, Vitaly Mikhailov⁽³⁾, Robert Emmerich⁽⁴⁾, Dmitrii Stoliarov⁽¹⁾, Ruben S. Luis⁽⁵⁾, Jiawei Luo⁽³⁾, Hideaki Furukawa⁵, Colja Schubert⁴, Ronald Freund⁴, David J. DiGiovanni³, Yuta Wakayama², Takehiro Tsuritani², Ian Phillips⁽¹⁾, Sergei K. Turitsyn⁽¹⁾

⁽¹⁾ Aston University, Aston Triangle, Birmingham B4 7ET, UK; a.donodin@aston.ac.uk

⁽²⁾ KDDI Research, Inc., 2-1-15 Ohara, Fujimino-shi, Saitama 356-8502, Japan;

⁽³⁾ Lightera Laboratories, 2000 Northeast Expressway, Somerset, NJ 08873, USA;

⁽⁴⁾ Fraunhofer Heinrich-Hertz-Institute, Einsteinufer 37, 10587 Berlin, Germany;

⁽⁵⁾ National Institute of Information and Communications Technology, 4-2-1 Nukui-Kitamachi, Koganei, Tokyo 184-8795, Japan.

Abstract We report a record GMI-estimated transmission of 172.87 Tb/s over 25 km of G.652.D in the O-band, achieved using 633×24.5 GBaud DP-64QAM and DP-128QAM channels, with optimized launched signal and Raman pump powers. ©2026 The Author(s)

Introduction

The rapid growth of data traffic driven by cloud services, artificial intelligence, and data centre interconnects continues to push optical communication systems towards higher capacity and improved energy efficiency. While ultra-wideband (UWB) transmission across multiple spectral bands has demonstrated record throughputs exceeding 400 Tb/s [1–3], it relies on complex multi-band amplification schemes also achievable under power constraints in deployed links [3]. In this context, the O-band (1260–1360 nm) has emerged as an attractive alternative for high-capacity transmission due to its wide bandwidth of 17.5 THz and near-zero chromatic dispersion in single-mode fibre (SMF), which enables simplified digital signal processing and reduced transceiver complexity [4, 5]. In addition, recent advances in bismuth-doped fibre amplifiers (BDFAs) have enabled simple, broadband amplification across the O-band, facilitating high-capacity coherent transmission without the need for multiple amplifier technologies [6]. They have enabled O-band throughput above 150 Tb/s to be demonstrated over tens of kilometers [5] using dense wavelength division multiplexing (DWDM). Nevertheless, O-band transmission remains challenged by higher fibre attenuation and enhanced nonlinear interactions. In particular, the low-dispersion regime increases the efficiency of four-wave mixing (FWM), leading to nonlinear interference (NLI) noise, which degrades the signal-to-noise ratio (SNR). Further capacity improvement requires careful optimisation of the optical signal-to-noise ratio (OSNR), including the management of wavelength-dependent loss, intra-band stimulated Raman scattering (SRS), and FWM.

In this work, we demonstrate a record 172.87 Tb/s GMI-estimated O-band coherent

transmission over 25 km of G.652.D fibre using 633×24.5 GBaud dual-polarisation (DP) 64QAM and 128QAM channels from 1266 nm to 1358 nm (16.14 THz). In addition, we demonstrate the highest average O-band spectral efficiency of 10.7 bit/s/Hz to date through optimisation of the launched signal and Raman pump at 1200 nm. We show that the performance is governed by the interplay between wavelength-dependent loss, SRS-induced power redistribution, and nonlinear interference noise. These results highlight the potential of O-band coherent systems as a high-capacity and energy-efficient solution for short-reach and data center interconnect applications.

Experimental Setup and Results

Figure 1 shows the experimental setup for the O-band transmission of 633×24.5 GBaud channels. Three independent external cavity lasers (ECLs) were used to generate 25.5 GHz-spaced data carriers. The central laser was used as a channel under test (CUT) and the other two were used as interfering neighbour dummy-channels. The CUT lightwave was sent into a 40 GHz dual-polarization (DP) IQ modulator, while the neighboring channel lightwaves were combined by a 3 dB polarization maintaining (PM) coupler. After combining they were amplified by a bismuth-doped fibre amplifier (BDFA) and a single linear polarization state was filtered out by a polarization beam splitter (PBS). This was done to prevent damage to the modulator by ensuring proper alignment of the signal with the operating axis of the DP-IQ modulator. After the PBS, the neighbour channels were modulated in a separate 40 GHz DP-IQ modulator. Both modulators were driven by a 120 GSa/s digital-to-analog converter (DAC). 64 QAM and 128 QAM signals were utilized for two different parts of the O-band.

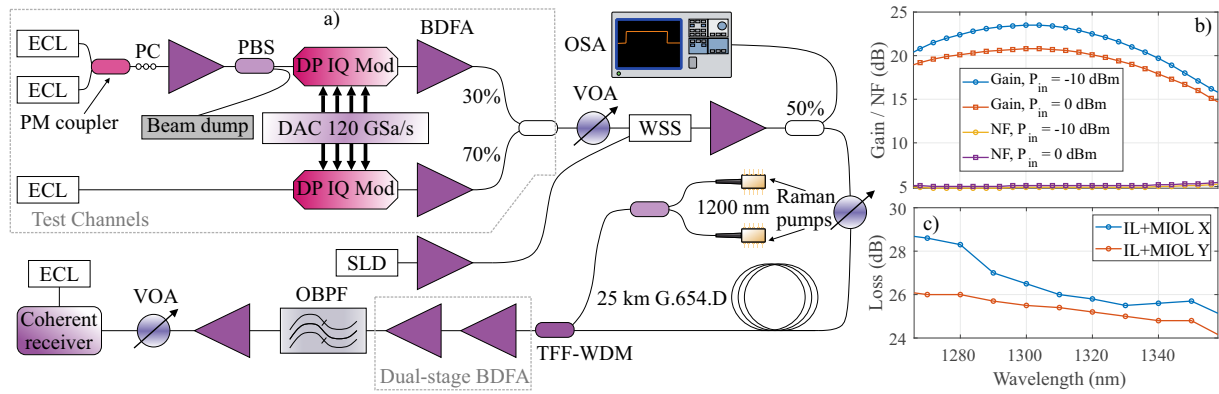


Fig. 1: a) Experimental setup for the transmission of O-band signal over 25 km of G.654.D; b) gain and NF of the BDFA; insertion loss (IL) and modulation induced optical loss (MIOL) of the CUT modulator for X and Y polarizations.

The input power to the modulator was between 11.5–13 dBm across the band for the CUT and between 17–20 dBm for the combined neighbor channels. Two individual BDFAs with an internal output power control were utilized to separately amplify the neighbor channels and the CUT. The three channels were then combined by a 30:70% coupler and a variable optical attenuator (VOA) was used to control the power of the test channels. Spectrally-shaped amplified spontaneous emission (SS-ASE) was used to emulate the fully-loaded O-band (1266–1358 nm). To create the SS-ASE, a superluminescent diode (SLD) and a BDFA were used and a 4 channel wavelength selective switch (WSS) was used to both shape the ASE and combine it with the test channels [7]. After the test channels were combined with the SS-ASE, the test band was amplified with another BDFA. A 3 dB wideband coupler was used to control the shape of the ASE and the power level of the test channels. A VOA was used to set the total launch power into the span to be 15 dBm with the $\sim$ 13dBm per channel launched power. The span comprised of 25 km-long G.654.D fibre with the span loss shown in Figure 2 (b). To compensate for fibre higher loss at shorter wavelengths, we utilized a single wavelength backward distributed Raman amplifier (DRA). The DRA comprised two Raman pumps operating at 1200 nm, combined using a polarization beam combiner (PBC) and coupled into the transmission fibre via a thin-film filter wavelength-division multiplexer (TFF-WDM),

with the pump port configured as the reflective channel. After the transmission, the signal was amplified by a dual-stage BDFA. An optical band-pass filter (OBPF) was used to select the CUT with another BDFA amplifying the CUT. The signal was received by an O-band coherent receiver based on 59 GHz 256 GSa/s optical oscilloscope. Another ECL was used as a local oscillator with the wavelength matching the central CUT of the test channels.

The BDFAs employed bismuth-doped fibres (BDFs) with various lengths in the range of 100 m to 220 m and were backward-pumped at 1150 nm with optical powers between 4 W and 5 W. Apart from the first stage of the double-stage BDFA, which was forward pumped at 1195 nm with pump power of 500 mW. All amplifiers used in the experiment apart from the dual-stage BDFA exhibited similar performance. The characterization of one single-stage BDFA is presented in figure 1 (b). The NF for both -10 and 0 dBm input signal powers were around 5 dB with gain between 15 to 20 dB for 0 dBm input signal power, and gain between 17 to 24 dB for -10 dBm input signal power.

In short-reach O-band transmission, low chromatic dispersion enhances four-wave mixing (FWM), which in multi-channel systems manifests as nonlinear interference (NLI) and appears as excess intensity noise, degrading the effective optical signal-to-noise ratio (OSNR). To estimate such impact the method of measuring notch depths has

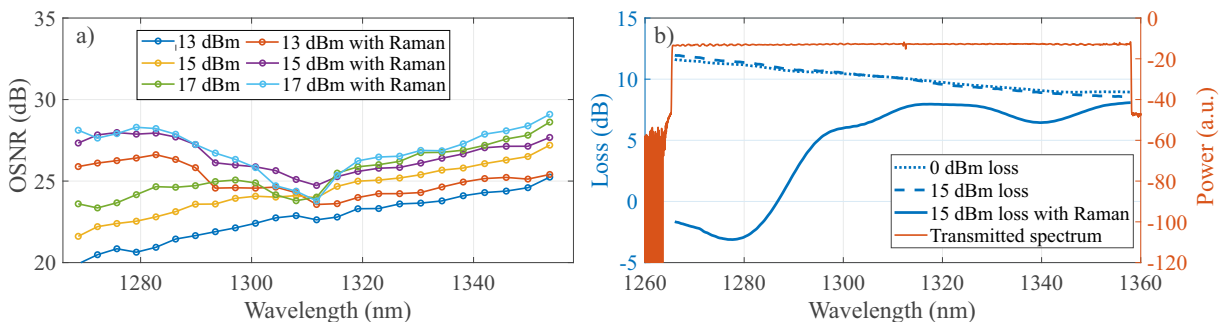


Fig. 2: a) Achievable OSNR for at the output of dual-stage BDFA for different input signal powers and Raman pump conditions; b) Spectra of the signal at the input of the span (red), loss profiles of span for different conditions (blue).

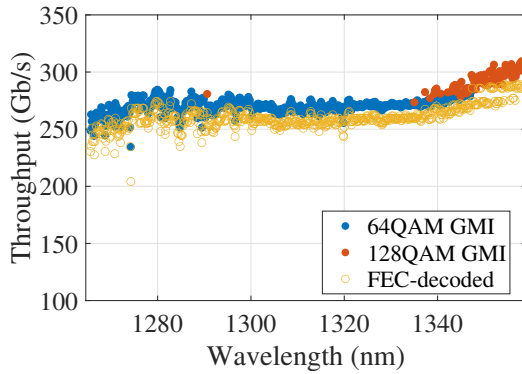


Fig. 3: Wavelength dependent data rate estimated using the GMI and after FEC decoding.

been proposed [8] and recently utilized for optimization of DWDM transmission in the O-band [3]. Here, we offer a supplementary correction to the method to also include the impact of the amplifier at the end of the line. This aspect is particularly important in systems with hybrid amplification and effects like intra-band stimulated-Raman scattering and FWM. Amplifiers are dynamic systems with typically non-uniform gain and noise penalty over the operation bandwidth, which can also strongly change with input signal powers.

To achieve optimal system performance, a series of experiments were conducted to evaluate the OSNR at the output of the pre-OBPF amplifier for different launched signal powers and Raman amplification conditions. Two Raman pump wavelengths, 1200 nm and 1240 nm, were investigated. However, the use of the longer-wavelength pump resulted in significant signal depletion in the 1266–1300 nm range, caused by strong intra-band SRS, as well as reduced gain in the in-line amplifier due to lower signal power.

A graphical summary of the achieved OSNR after transmission before the OPBF under different launching conditions is presented in Fig. 2. The reduced OSNR at shorter wavelengths, observed for all launch powers after 25 km transmission, is attributed to increased span loss, leading to lower input signal power to the amplifier. The onset of significant FWM impact is observed at a launch power of 17 dBm, evidenced by a noticeable dip in the received OSNR at 1306 nm, corresponding to the zero dispersion wavelength.

The degradation of OSNR within the amplifier can be substantially mitigated through the use of distributed Raman amplification (DRA) across all launch powers. While both 17 dBm and 15 dBm launch powers exhibit comparable overall performance, 17 dBm provides slightly improved OSNR at longer wavelengths. In contrast, 15 dBm combined with Raman amplification enables a more uniform OSNR profile across the band, exceeding 25 dB over the entire spectrum. Based on these results, a launch power of 15 dBm was selected.

The launched spectrum is shown in Fig. 2(b) in red, where the channel under test (CUT) and neighbouring channels are visible around 1310 nm. The span loss is also presented in Fig. 2(b): the blue dotted line corresponds to a 0 dBm launched signal power, while the blue dashed line corresponds to 15 dBm. The more pronounced tilt in the latter indicates the onset of intra-band SRS, which increases the effective loss at shorter wavelengths and reduces it at longer wavelengths. The net span loss with Raman pumps enabled is shown by the solid blue curve. A slightly negative loss is observed in the 1266–1290 nm range, indicating Raman overcompensation of the fibre loss. Raman gain is evident across the entire O-band, enabled by partial pump amplification and enhanced intra-band Stimulated Raman Scattering (SRS), which promotes power transfer from shorter to longer wavelengths.

The system performance was evaluated by estimating the achievable data rate using both the generalized mutual information (GMI) and post-FEC decoding. For the latter, we adopted the approach used in [1], employing FEC codes from the DVB-S2 standard [9] with puncturing to provide finer code-rate granularity. Code rates were evaluated until a bit-error rate below 10^{-5} was achieved, and a 1% overhead hard-decision code was assumed to eliminate residual errors [10].

Figure 3 presents the wavelength-dependent data rate, obtained by averaging five 10 μ s acquisitions, corresponding to a total of $1.2 \cdot 10^6$ symbols. For each channel, the modulation format was selected to maximise throughput. The per-channel data rates show negligible impact of FWM on the transmitted signal. The highest performance is observed at longer wavelengths, where modulator impairments are reduced within the considered range, indicating potential for further improvement with an optimised O-band transmitter. A total throughput of 172.87 Tb/s was estimated based on GMI, with 164.83 Tb/s achieved after FEC decoding, corresponding to an implementation penalty of 4.6%. These results highlight the strong potential of O-band transmission for high-capacity systems.

Conclusions

We achieved the highest transmission throughput reported within the O-band, reaching a GMI-estimated data rate of 172.87 Tb/s and an FEC-decoded data rate of 164.83 Tb/s over a spectral range of 16.1 THz. An average spectral efficiency of 10.2 bits/s/Hz was also obtained, representing the highest reported post transmission value in the O-band. We further demonstrate that, through launch power optimisation and the careful application of distributed Raman amplification, low FWM impairments can be maintained while preserving high throughput across the entire O-band.

Acknowledgements

AD was supported by the DSIT and the RAEng under Research Fellowship. This work was supported by Royal Society grant IEC/R3/243193, UK EPSRC grant TRANSNET (EP/R035342/1) and grant EP/W002868/1, parts of the work - by the German Federal Ministry of Research, Technology and Space (BMFTR) under HYPERCORE (grant ID 16KIS2097K).

References

- [1] B. J. Puttnam et al., “402 Tb/s GMI data-rate OESCLU-band transmission”, in *Optical Fiber Communication Conference*, Optica Publishing Group, 2024, Th4A–3.
- [2] R. Luis et al., “430 tb/s GMI data rate over a standard G. 654 fiber using few-mode O-band and single-mode ESCL-band transmission”, in *2025 European Conference on Optical Communications (ECOC)*, IEEE, 2025, pp. 1–4.
- [3] R. S. Luis et al., “450 Tb/s GMI, 42.4 THz, OESCL-Band Transmission Over a Field-Deployed Fiber”, in *Optical Fiber Communications Conference*, Optica, 2026, Th4B.5.
- [4] D. J. Elson et al., “Continuous 16.4-THz bandwidth coherent DWDM transmission in O-band using a single fibre amplifier system”, *Journal of Lightwave Technology*, vol. 43, no. 4, pp. 1813–1818, 2024.
- [5] R. Aparecido et al., “153.8 Tb/s O-Band Coherent Transmission over SMF with Low-Complexity DSP”, in *Optical Fiber Communication Conference (OFC)*, Optica, 2026, W3J.4.
- [6] V. Mikhailov et al., “1255–1355 nm (17.6 THz) bandwidth O-band BDFA pumped using uncooled multimode 915 nm laser diode via YDF conversion stage”, *Journal of Lightwave Technology*, vol. 42, no. 4, pp. 1265–1271, 2023.
- [7] D. J. Elson et al., “Investigation of bandwidth loading in optical fibre transmission using amplified spontaneous emission noise”, *Optics express*, vol. 25, no. 16, pp. 19 529–19 537, 2017.
- [8] J.-X. Cai, O. V. Sinkin, D. G. Foursa, G. Mohs, and A. Pilipetskii, “Direct measurement of broadband FWM induced noise in dispersion uncompensated systems”, in *2013 Optical Fiber Communication Conference and Exposition and the National Fiber Optic Engineers Conference (OFC/NFOEC)*, IEEE, 2013, pp. 1–3.
- [9] D. Second Generation, “Digital Video Broadcasting (DVB); Second Generation”, 2013.
- [10] D. S. Millar et al., “Design of a 1 Tb/s superchannel coherent receiver”, *Journal of Lightwave Technology*, vol. 34, no. 6, pp. 1453–1463, 2016.